A proof of the conjectured order of the recurrence for OEIS A268454

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1 The conjecture

The Online Encyclopedia of Integer Sequences records A268454 as

> *Number of length-n 0..5 arrays with no adjacent pair x,x+1 followed at any distance by x+1,x*

and carries in its Formula section the line:

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Empirical recurrence of order 51 (see link above)
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That line carries no separate signature; the entry itself is by R. H. Hardin (Feb 04 2016).

The word *Empirical* — equivalently *Conjecture* — is the entry's own: the statement is recorded as fitted to the published terms, and no proof is given there. As of revision 4, last modified Feb 04 2016, the entry records no proof and links to none, so the statement is open. This note settles it.

The entry states only that the sequence satisfies a linear recurrence of order 51 and refers to a linked file for its coefficients, which this note does not use. What is proved below is the corresponding exact statement: the least order of a linear recurrence the sequence eventually satisfies, and the exact index beyond which it holds.

2 The object

The name is read as follows. The reading is not a guess: it is pinned against the entry's own published terms, and against an independent enumeration, before anything is proved (Section 7).

The objects are the words $w_1 \cdots w_L$ over $\{0, \ldots, 5\}$ of length $L = n$. Call w_i a *repeated value* when $w_i = w_{i-1}$, and read the repeated values of a word in the order in which they occur.

The entry's condition is: *no adjacent pair x,x+1 followed at any distance by x+1,x.*

$a(n)$ is the number of such arrays. The entry's offset is 1, so its term of index n is the count for n rows.

3 The count is a walk count

Read the word one letter at a time. Whether the new letter is a repeated value is decided by the letter before it, and the condition compares it with a bounded amount of information about the repeated values already seen. Taking that information together with the last letter as the state makes the whole condition decidable one letter at a time.

A word of length L is then a walk of length L from the empty state, and every word arises from exactly one walk, so $a(n) = w^{\mathsf{T}} M^{L} f$ and the count is C-finite.

4 The degree bound, derived

The reachable part of the digraph has 193 states, 193 after trimming; the forward identification of states with equal numbers of completions of every length, followed by the mirror identification on the edges coming in, leaves $S = 156$ blocks. The count satisfies the linear recurrence given by the characteristic polynomial of that 156×156 matrix, and that bound is computed, not assumed.

5 The decision procedure

Let $c_1, \ldots, c_D$ be the coefficients of a claimed recurrence $a(n) = \sum_{i=1}^{D} c_i\, a(n-i)$, let $q(t) = t^D - \sum_{i=1}^{D} c_i t^{D-i}$ be its characteristic polynomial, and put

$$u_m \;=\; \tilde{w}^{\mathsf{T}} L^{m-1} q(L)\, \tilde{f} \qquad (m \geq 1).$$

Expanding the powers of L and using the displayed formula for a, one gets $u_m = a(n) - \sum_{i=1}^{D} c_i\, a(n-i)$ with $n = m + D$. So u_m is exactly the residual of the claim at that index, and the recurrence holds at an index n if and only if the corresponding u_m vanishes.

Now $u_m = \tilde{w}^{\mathsf{T}} L^{m-1} y$ with $y = q(L)\tilde{f}$ a fixed vector, so (u_m) satisfies the characteristic polynomial of L, of degree 156: writing that polynomial as $t^{156} - \sum_{i=1}^{156} \lambda_i t^{156-i}$ gives $u_{m+156} = \sum_{i=1}^{156} \lambda_i u_{m+156-i}$. Hence 156 consecutive vanishing residuals force every later residual to vanish, by induction. The test is therefore finite; and because it locates the *last* nonvanishing residual, it pins the threshold exactly rather than merely bounding it.

Everything is carried out in exact integer arithmetic on the vector $L^{m-1}y$. The matrix M is never formed.

6 Results

Theorem 1. *For A268454, the least order of a linear recurrence that a eventually satisfies is exactly* 51*, and that recurrence holds for every* $n > 51$*, the entry's own figure.*

Proof. The count is C-finite of order at most $S = 156$, so the minimal annihilator of the tail $(a(n))_{n \geq M}$ is the same for every M beyond S: the nilpotent part of L is killed after S steps. Taking such an M and 318 consecutive terms from there, the Berlekamp–Massey algorithm over $\mathbb{Q}$ returns the minimal linear recurrence generating them; since the sequence satisfies one of order at most S, $2S$ terms determine it. Its order is 51. That recurrence was then put through the residual test of Section 5: every residual vanishes, followed by more than $S = 156$ consecutive vanishing ones, so it holds for every larger n. No recurrence of smaller order can hold eventually, since the tail's minimal annihilator is what the algorithm returned. $\square$

7 Verification

Four checks were run, each reproducible from the code accompanying this note, and the first two are what pin the reading of the entry's English.

- **The published terms.** The walk count reproduces all 20 terms the entry publishes, exactly, in integer arithmetic, with the index taken from the entry's stated offset (1) rather than fitted. The first of them are 6, 36, 211, 1231, 7160, 41526, 240170, 1385338, ...
- **An independent enumeration.** For $n = 1, \ldots, 6$ every one of the 6^{1n} arrays was written out and tested cell by cell against the condition as the entry states it — no transfer matrix, no row window, a separate piece of code. The counts agree with the walk count and with the entry.
- **The claim on the raw data.** Each claim was evaluated on the entry's published terms alone, with no model involved, wherever the proved range and the published data overlap. It holds there.
- **The annihilation test.** Carried to the derived bound $S = 156$ over the whole vector, in exact integer arithmetic, never sampled and never truncated.

8 References

1. OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A268454, revision 4, last modified Feb 04 2016.
2. R. P. Stanley, *Enumerative Combinatorics*, Vol. 1, 2nd ed., Cambridge University Press, 2012; §4.7, the transfer-matrix method.
3. M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011; C-finite sequences and their closure properties.
4. J. Berstel and C. Reutenauer, *Noncommutative Rational Series with Applications*, Cambridge University Press, 2011; linear representations of counting automata and their reduction.